Two-Tiered Unionization and the Local Labor Market *

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Abstract

What makes workers organize themselves in a context where free-riding is easy? I investigate drivers of unionization in the Swedish labor market, characterized by near-universal collective bargaining agreement coverage and strong sectoral unions. Workers can opt for plant-level unionization in order to adapt national agreements to the local level. I develop a model where the level of bargaining is an endogenous choice on the worker side to extract theoretical predictions with regards to why plant-level and sectoral bargaining yield different resource allocations, and how worker preferences over these levels shift. I show that local unionization is more flexible in adapting to local circumstances, and serves as a mode of insurance in economic downturns. It follows that local unionization should be a decreasing function of employed workers' outside options. I confirm this prediction using data on plant-level union chapters combined with administrative data. Utilizing notices of mass layoffs within the local labor markets as negative shocks to incumbent workers, I employ a matched difference-in-differences design to find that unionization increases at surrounding plants, a result stemming from more union members among incumbents. Local union chapters serve as a mechanism through which workers can effectively insure themselves, and their propensity to utilize that tool depends on the value of their outside options.

Keywords: Unionization, Collective Bargaining, Outside options .
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1 Introduction

Collective bargaining agreements (CBA:s) are a key feature in most developed countries, and are strongly tied to worker unionization. However, the sectoral nature of many agreements imply that the share of covered workers often supersede the share of unionized workers. Indeed, the OECD average of CBA coverage was measured to 34% in 2024, compared to an average unionization rate of 15% (OECD 2025). While the worker bargaining parties explicitly derives their strength from union membership, an individual worker often has little incentives to spend time and money unionizing themselves. This is a classic free-rider problem: as CBA:s are extended to non-unionized members, the contents function as a public good where the worker can reap the benefits without personally investing. Yet, workers do unionize, even in contexts where CBA:s are mandated and universal.

Moreover, many countries differentiate between sectoral, firm-level or plant-level collective bargaining. The question of welfare optimal bargaining tiers goes back to at least Calmfors and Driffill (1988), who argue that the relation between bargaining centralization and macroeconomic performance follows a hump-shaped pattern, where the optima are reached either through plant-level bargaining or national bargaining. Less, however, is known about worker preferences over these vertical tiers. For centralization to matter in macroeconomic terms, it must be the case that allocations of resources differ by vertical tier. It then follows that workers are directly affected by the level of bargaining as these allocations enter their utility functions.

In this paper, I study these phenomena through two research questions. Firstly, why and how do the resulting resource allocations differ by the vertical level of bargaining? Secondly, how do worker preferences over these tiers depend on the workers' status on the labor market?

To answer these question, I first set up a search and match model with match-level and national tiers of bargaining. Central to the model is that separations are endogenous, unilateral decisions taken by the firm at a separation cost, which itself is a bargaining object along with wages. Key is then the tradeoff between employment security and wages, with optimal allocations differing over degrees of centralization.

I thereafter conduct an empirical analysis in the Swedish setting. Using rich administrative data combined with novel data collected directly from the unions on plant-level organization and CBA coverage, I can characterize unionization decisions and an expansive set of possible covariates. To investigate how unionization responds to changes in the local labor market, I utilize notices of mass layoffs at a nearby plant within the same industry as treatment in a matched difference-in-differences design. The key idea is that workers could potentially respond to changes in outside options through the local unionization mechanism. Specifically, I posit that worsened prospects should workers get displaced affects their willingness to pay for protecting their current employment.

The key output from the model is that plant-level and central bargaining differ in their willingness to pay for employment security, with the former having stronger incentives to protect incumbent workers. This stems from two distinctions in their valuations. Firstly, local atomistic unions do not internalize externalities in that higher employment protection affects firm decisions on the hiring margin. The intuition is straightforward in that making layoffs harder will induce more selectivity in hiring decisions, spilling over to the unemployment rate. Secondly, I posit that the central unions explicitly care about the employment rate, and thus take these spillovers into account. As the resulting allocations imply more employment protection with local unionization, it is de facto a mode of insurance. Since the bargaining tier is a worker-side unilateral choice, I find that they are more willing to unionize locally when the value of insurance rises, i.e. when times are tough for the unemployed.

Descriptively, there is a strong positive correlation between unionization and local unemployment rates. This is consistent with the model in two dimensions: while local unionization can drive up unemployment rates, weak labor markets themselves induce unionization as tools for incumbents to protect their employment. Looking directly at mass layoffs, it is further the case that these local unions do succeed in protecting employment, as layoffs occur as frequently as at non-unionized plants but result in significantly higher employment retention. Furthermore, unionized plants exhibit lower action on the hiring margin both in terms of net hiring and new employees, consistent with the modeling implication that higher separation costs induce more selectivity on the hiring margin.

To remedy the reverse causality issue between unionization and unemployment, I shock the outside option of employed workers with a mass layoff within the local labor market. I find that worsening the alternative prospects induces a significant increase in union density as compared to a matched control plant within the same industry. This effect is driven entirely by the number of union members at the plant, as opposed to a differential restructuring of the workforce. I do not see any effects of the treatment on the extensive margin of plant-level union formation.

Broadly speaking, this paper fits into three strands of literature. It firstly contributes to the literature on unionization and its effects by considering a case where the gains are not ex ante obvious due to the free-riding problem of near-universal collective bargaining agreement coverage. Secondly, I contribute to the literature on worker sensitivity to changes in outside options, by highlighting unionization as a mechanism for workers to act upon changing alternative prospects. Thirdly, it contributes to a theoretical literature on frictional labor markets by highlighting how a two-tiered union system interacts with labor market structures.

The idea that unions do not spring to life by themselves, but instead are induced by labor market conditions, is not a new one. Already Lewis (1963) notes that relative wages between unionized and non-unionized sectors might shift people into unionization, a sentiment echoed by e.g. Reder (1965) and Ashenfelter and Johnson (1972). This relates to a larger strand of literature focusing on the so-called union wage premium, an increase in wages for a unionized worker relative to a non-unionized worker in the same occupation gained through bargaining as a collective. Notably, there are many countries where non-unionized workers are covered by CBA:s such that the scope for wage premia should be limited. While e.g. Budd and Na (2000) and Hildreth (2000) argue that premia can exist even within CBA coverage, Booth and Bryan (2004) find that taking the endogeneity into account eliminates the effect. Notably, net gains to unionization are plausibly heterogeneous: Acemoglu et al. 2001 argues that the top end intuitively has less incentives to compress wage distributions. Moreover, firms can take this into account. Taschereau-Dumouchel (2020) finds that the threat of unionization makes firms adjust their workforces as to avoid collective bargaining. While inefficient input allocations implies welfare losses, this can be seen as a counteracting force to monopsonistic employment

practices. Benmelech et al. (2022) find that monopsonistic markets have lower wage levels, but that this effect is smaller for more unionized industries. This is further studied by Azkarate-Askasua and Zerecero (2025), who estimate a general equilibrium model in France and find that unions reduce the welfare losses induced by monopsony.

This paper further contributes to the literature on how changes in workers' outside options affect them when they are already employed. The theoretical intuition is simple: if an employed worker's prospects on the labor market increases, their loss in surplus from separating from their current job decreases. This argument is further consistent with the literature on unemployment insurance, which similarly raises the value of unemployment. A large part of this literature focuses on wage effects. Caldwell and Harmon (2019) use Danish data on coworker networks to instrument for outside options, and find that increases in outside options positively affect wage growth as well as job-to-job transition rates. They argue that the wage growth mechanism stems from re-bargaining wages once learning about their strength on the labor market. This is consistent with Jäger et al. (2024), who use a survey in Germany to elicit worker beliefs about outside options and supply workers with true values of comparable wages, and find that those that underestimated their strength report higher probabilities to renegotiate wages, as well as look for a new job or quit the current one. I contribute here by introducing a theoretical mechanism through which worsening of outside options indeed reduces wages, but not through a direct spillover in the wage-setting process. Instead, I consider the effect to come through increased worker propensity to trade away wage growth for increased job security. The closest paper in this context is Pezold et al. (2023), who use a survey in the U.S. context to investigate how beliefs about labor market tightness affect unionization decisions, and find significant but economically small procyclical effects. Their proposed mechanism comes through employer retaliation: I instead consider the choice of bargaining level as the margin of response.

The paper finally speaks to a literature on frictional labor markets from a theoretical point of view. In section 3, I present a model of two-tiered unionization on a frictional labor market, where workers and firms maximize expected surplus over match lengths through trading off wages and firm separation costs. The backbone of the model is Pissarides (1985), which constitutes a canonical DMP model with heterogeneity in match

productivity. In addition to the multi-tiered bargaining, I add two other alterations to the model: I let match productivity be time-variant, and I make all separations endogenous as opposed to matches being destroyed through some exogenous parameter.

These additions are naturally connected - in order for a match to be endogenously destroyed, something has to happen within the spell for the parties to separate. This has been done previously in Mortensen and Pissarides (1994), where match surplus is adjusted through aggregate productivity shocks arriving through an exogenous process and affecting prices for all firms. In that model, separations are costless and occur when the shocks are such that surplus is non-negative. I depart from that model through having the variation occur within-match instead of in the aggregate, thus letting match productivity be time-variant as opposed to prices, and further allow for non-zero separation costs.

Notably, an implication is that termination will be a unilateral decision on the firm side, and not be a termination by mutual agreement. This is in contrast with e.g. Fujita and Ramey (2012), where separations are endogenous but occur when it is optimal for both parties. More closely to my approach is Carlsson and Westermark (2022), where separation decisions are not jointly optimal, and stem from both aggregate and idiosyncratic productivity fluctuations. The key idea in that paper is that wages are sticky on the incumbent margin, leading to effects on unemployment. This will prove to be true in my setting as well, whereas I introduce local union bargaining as a mechanism for being more flexible in wage moderation. The idea that wage rigidity implies inefficient separations is further echoed in Davis and Krolikowski (2025), who employ a survey to find that among UI recipients, a majority would have accepted a 5% reduction in wages in order to keep their previous employment, in contrast to the efficient separation hypothesis.

The closest paper in terms of modeling is Blanco et al. (2024), where worker productivity evolves stochastically and there is a two-sided lack of commitment such that both layoffs and quits occur endogenously due to wage rigidity. A key difference between the models is that their model has directed search, wage posting, and worker-specific productivity. Interestingly, the downstream results with regards to search frictions turn out to be qualitatively similar as their search setup mirrors a Nash bargaining problem for wages. However, my choice to make match productivities idiosyncratic instead of individual prohibits me from drawing macroeconomic conclusions about the productivity distribution

on the aggregate, which is not the focus of the paper. I further limit myself to firm-side unilateral separations: this is to make the model more tractable, and has a basis in the empirical observation that $E \to U$ quits are rare.

The paper is structured as follows: I firstly describe the Swedish institutional setting, which will be used to conduct empirical analysis and motivate modeling choices. Section 3 presents the theoretical framework to understand the mechanisms at play, and extract theoretical predictions. In the next section, I describe the data and estimation strategy used to test these predictions. I follow up with descriptive and causal empirical results and conclude with a discussion.

2 Institutional Setting

2.1 Overview

Sweden, along with its Nordic neighbors, are among the most unionized countries in the world (OECD 2025), and roughly 80% of Swedish workers are covered by Collective Bargaining Agreements (CBA:s). However, looking at the trend, Swedish unions have been facing a declining membership rate since 2006 (Kjellberg 2019). This is especially pronounced within the blue-collar labor market as they have been surpassed by their white-collar counterparts in coverage, even though they have historically been dominating.

A characterizing mark of the Swedish model is the lack of interference from the government, which has been a core feature since its implementation in 1938 (Uusitalo et al. 2025). As noted by e.g. (Levinson 1997), the central labor market parties, i.e. the employer confederations and central unions, exhibit a large degree of cooperation even though they serve as adversaries in negotiations. From a game theoretical point of view, these negotiations can be thought of as a repeated game where uncooperative equilibria would lead to government involvement. As neither party desires that outcome, they have built up mutual trust to maintain cooperative equilibria. This is in line with the empirical observation that it is rare with strike events, as disputes typically are settled before reaching that stage, and forbidden during an ongoing CBA.

There are however parts of the codetermination in the labor market that are codified in law. Notably, MBL, the law on *Co-determination in the Working Life*, provides legal guidelines for when unions must be consulted, or bargained with, in employer decisions (Riksdagen 1976). This includes, but is not limited to, restructuring, layoffs, recruitment, investments and relocation. Moreover, the Swedish EPL is highly relevant to co-determination. The EPL specifies a last-in-first-out (LIFO) system that is to be applied in layoffs, where generally the worker with the lowest tenure for a given occupation is to be laid off first. This pertains to unions in that they can negotiate LIFO away if they wish, letting employers be more selective in their layoffs. Through MBL and LIFO, the union thus has a significant amount of power in making layoffs easier or harder for the firm. Key to interpreting is that these provisions are mostly dispositive: as the rules

are applied by default, they consist the outside option in any bargaining procedure, and the more rigid the rules, the more potential gains from trade at the local level.

2.2 The two-tier system

Swedish unions are two-tiered. On the central tier, there are a total of 57 unions, with each union covering a collection of occupations, sometimes split by sector or whether or not the employer is private. They are organized in three larger umbrellas: LO (Blue-collar), TCO (White-collar), and SACO (Academics), who are roughly equal in membership size. These central unions then have local chapters, known as *clubs*, that organizes workers at a specific plant. Forming a local club de facto transfers most MBL bargaining mandates from the central to the local level. That is, most bargaining outcomes reached by a firm and a central union can be circumvented by the club and the local employer agreeing on something else. This implies that the central union serves as a form of outside option for workers organized in a club. While not all central agreements can be undone locally, bargaining mandates related to layoffs are universally transferred to clubs.

The formation of local clubs does not, as opposed to e.g. the U.S. context, require a vote such that the majority of workers at a plant support the proposition. Instead, it only requires a few workers (typically 3-5) belonging to the same central union to form the local chapter. These local chapters thereafter have internal democratic elections to delegate e.g. chairmanship and negotiation rights. The clubs are formally connected to the central unions, and can be thought of as plant-level branches. This implies that they enjoy the benefits of being connected to a large apparatus, e.g. through having lawyers and other experts at the ready, without having to fully conform with the sometimes rigid CBA provisions. Formation is fully unilateral in that the employer has no say in the process. Notably, however, is that the local employers are positive towards plant-level unionization (Levinson 1997) as it allows them to adjust the centrally bargained outcomes to the local reality. This contrasts e.g. Dustmann et al. (2025) where firms in Italy in order to be more flexible choose to leave the CBA system entirely due to its rigidity - here the institutional context allows for flexibility within the system.

2.3 Collective bargaining

Notably, the Swedish union system is such that any CBA is negotiated at the firm-sector level, and that all individuals within a signatory firm are covered regardless of their union status. This naturally creates free-riding issues as union membership comes at a monetary cost, whereas key benefits of membership are universal. There are still however individual gains to membership. Depending on the union, these range from e.g. legal assistance, supplemental unemployment insurance, job search assistance, and discounts on consumption goods such as home insurance. That being said, most of the unions' output come through the collective bargaining channel, central or local. Thus, employed workers at a plant rather face the decision of whether or not they prefer the central or local bargaining outcome (and process).

The contents of CBA:s are very heterogeneous, especially when it comes to wage bargaining. As a general norm, Swedish wage formation is done through pattern bargaining, where the export-oriented manufacturing sector is the first to set wages. The rest of the unions then use this outcome as a benchmark in their separate agreements, either informally or through legal anchoring. While through the pattern bargaining process the general level of wage increases is decided centrally, the distribution of these increases is sometimes left to the local level depending on the CBA. For the most centralized CBA:s, there is no scope for local bargaining as all wages are specified, whereas the least centralized leave all bargaining to the individual, disregarding the pattern bargaining outcome. Most of the workforce fall somewhere in the middle, with e.g. centrally supplied average increases but individual level bargaining, sometimes with a pre-specified outside option. Moreover, these CBA:s contain non-wage provisions regarding e.g. working hours, vacation days, retirement provisions and mandatory notice.

2.4 The notice procedure

Due to EPL, all layoffs such that a firm aims to separate from at least 5 workers during a short time frame have to be reported to the PES. This report contains information on how many employees they aim to lay off, as well as why. This is the de jure start of a bargaining procedure where the employer and the union, either locally or through a

central representative, try to agree on the extent of the layoff, as well as on how the LIFO-circuits should be constructed. After the bargaining has concluded, the plant reports a finalized list of employees to be laid off.

As described above, unions can have significant impacts on the outcome, e.g. locally through circumventing LIFO or centrally through e.g. mandatory notice times. A possible outcome of the bargaining procedure is further a voluntary severance pay package, where individual workers can choose to forego their mandatory notice period in exchange for monetary compensation.

3 Model

In this section, I develop a DMP-type model of two-tiered bargaining with endogenous separation to highlight theoretically why local and central unions differ in their objectives, and why workers would prefer one option over the other. In short, local union clubs by construction consist of workers *already employed* - as such they have less incentives to value the hiring margin than a central union who considers the entire economy. While both layers of unions maximize over a wage-employment security tradeoff, the local clubs value the latter relatively more. Moreover, the relative weight of employment security depends on the outside option such that preferences for employment protection, and hence for local clubs, increase when e.g. the unemployment rate rises.

The key idea of the model is that matches are unilaterally destroyed by firms due to changes in match productivity, which constitutes a loss of surplus for the workers. These separations come at a firing cost for the firms, which then generates value for the incumbent workers. These firing costs are bargained at the central level and take into account spillovers on the hiring margin and unemployment rate. Internalizing these externalities generates a suboptimal allocation for the employed workers, who can choose to unionize locally and assume the bargaining mandate, effectively trading off wages for higher firing costs. The returns, and incentives, for doing so is a monotonic function of labor market tightness, leading to the prediction that local unionization should rise with unemployment rates.

3.1 Setup

Consider an economy of unit mass where a fraction u are unemployed, and $1-u$ working. The unemployed are actively and inelastically searching for vacancies v, but there are frictions on the labor market. Defining the ratio of vacancies to unemployed as $\theta := \frac{v}{u}$, the workers encounter vacancies with flow rate $p(\theta)$, and vice versa, vacancies encounter candidates with rate $q(\theta)$. This implies $up(\theta) = vq(\theta)$ or equivalently $p(\theta) = \theta q(\theta)$. Time is continuous and there is a discount rate r.

Workers are in one of two states: employed or unemployed. When employed they generate productivity y without any labor elasticity, and in return get wages w which enter

linearly into their flow utility $u(w) = w$. When unemployed, they get home production value b and face the probability of finding a new match, depending on the probability of meeting a firm p and the probability of the match being satisfactory for both parties a.

There is a continuum of firms who either have an open vacancy or a filled position. There is no mass of workers at a firm, such that a firm is equivalent to a job. The product market is not modeled - all output is sold at the numeraire price $p = 1$ such that their flow profit equals $\pi = y - w$. Firms have several choices available to them depending on their state. If they have no filled position, they can choose to open a vacancy or not. An open position comes at the vacancy cost $c > 0$. Opening a position allows for meeting unemployed workers, which they can agree to form a match with, or separate from and return to the search. Once filled, the firm can at any point unilaterally separate from the worker at present-valued cost κ. There are no exogenous separations, such that all separations come from the firms' choices to lay people off. There is further a free entry condition such that firms will continue to enter the market as long as there is a positive value in doing so.

Productivity is match-specific and drawn from a continuous and differentiable distribution $F(y)$ each time an unemployed worker meets a vacancy. During a match, this productivity changes exogenously during the spell with shock distribution $G(z)$ and arrive at a Poisson rate of λ. The distribution and arrival rate are known to workers and firms. Productivity is always public information and as such observed by both parties throughout the spell. There is an endogenous productivity threshold $\tilde{y}$ such that no matches are formed if the initial $y < \tilde{y}$. Moreover, there is an endogenous separation productivity threshold x such that when productivity due to shocks fall below x, the firm will unilaterally destroy the match.

There are three tiers of bargaining in the model. Firstly, there's wage bargaining at the match level. Bargaining is done through a Nash protocol with worker bargaining power β over the expected generated surplus during the employment spell, which will have the endogenous length τ. The wages are fixed during the length of the match, and this rigidity will generate bilaterally inefficient separations as opposed to canonical models who typically find them jointly optimal. Secondly, there's Nash bargaining at the central level over the level of employment protection κ, where the central union has bargaining

power γ. The central level allows the union to take into account the effects changes in κ will have on e.g. wages and the unemployment rate. The local actors treat this κ as a fixed parameter. In a final stage, I allow local actors to renegotiate the value of κ, taking the centrally supplied value as an initial allocation.

3.2 Match-level bargaining

In mathematical notation, the value functions for the the non-filled states in the economy can be summarized by:

$$rV = -c + a(\tilde{y})q(\theta)[J^e - V] \tag{1}$$

$$rU = b + a(\tilde{y})p(\theta)[W^e - U] \tag{2}$$

Value is thus derived from expected transition gain, with J^e and W^e denoting the unconditional expectations over the initial and dynamic productivities of a filled position for firms and workers.

In the filled states, value is derived from production and wages for the length of the contract, i.e. as long as the firm doesn't choose to lay the worker off. The functions are most easily expressed as a system of variational inequalities with the state variable y.

$$rJ(y) = \max\left\{y - w + \lambda\Big[\int \max\{J(z), -\kappa\}dG(z) - J(y)\Big], -r(V+\kappa)\right\} \tag{3}$$

$$rW(y) = w + \lambda\Big[\int^x UdG(z) + \int_x W(z)dG(z) - W(y)\Big] \forall \quad y \geq x \tag{4}$$

The firms are after a realized shock in one of two zones. If the resulting productivity is above the separation threshold, i.e. if $y \geq x$, the match continues and the employer enjoys flow profits $y - w$ taking into account future productivity fluctuations. If the productivity drops below the threshold, they prefer paying κ to destroy the match, implying a value of $-r\kappa$ after taking the free entry condition into account. This further caps potential losses in the continuation zone. The workers have no actions to take but simply take their productivity into account through evaluating their future separation risk. Should

the productivity already have fallen below x in the realization of the shock, they have transferred to the unemployed state already.

From firm value 3 follows a simple firing rule in that as soon as productivity and its continuation value reaches a threshold, the firm prefers paying κ in order to move to the vacant state. Already here is clearly visible the value of κ to the employed worker, in that the firm is less likely to fire them if it is more expensive. To solve for the endogenous separation productivity, note that by definition it must satisfy $J(x) = -\kappa$, and that the firm's value function in the continuation zone can be rewritten as:

$$(\lambda + r)J(y) = y - w + \lambda \int_x J(z)dG(z) - \lambda G(x)\kappa \tag{5}$$

The implication is that $J'(y) = \frac{1}{r+\lambda}$: together with the boundary condition it follows that:

$$J(y) = \frac{y - x}{r + \lambda} - \kappa \tag{6}$$

Entering this expression into 5 gives the straightforward expression:

$$x = w - r\kappa - \frac{\lambda}{r + \lambda} \int_x z - x \; dG(z) \tag{7}$$

While not a closed form solution, the implicit function theorem can be used to show that the derivatives are signed in an intuitive manner. First, the positive relationship with w implies that a firm has higher productivity demands for well-paid workers. Second, higher firing costs lowers the productivity threshold such that the employer is more reluctant to let workers go. With the firing productivity, it is possible to characterize the wages through Nash bargaining:

$$w(y_0) = \arg\max_w (W(y_0) - U)^\beta (J(y_0) - V)^{1-\beta} \tag{8}$$

$$= \beta(y_0 - x(y_0) - (r + \lambda)\kappa) + rU \tag{9}$$

There are two counteracting forces in the expression. While productivity has a direct effect on wages through β, we know from (7) that higher wages imply higher firing thresholds,

which in turn pulls down the wage itself. Thus, x will be a direct function of y_0. Moreover, the worker implicitly pays for subsequent shocks and the future firing cost the firm will have to pay.

The wage expression can be used to find a hiring threshold productivity $\tilde{y}$. Naturally, the worker is indifferent between the states when $w(\tilde{y}) = rU$. Through the wage expression, this productivity level maps directly to an individual firing productivity, $x(\tilde{y})$. Rearranging (6) and setting it to zero then yields:

$$\tilde{y} = x(\tilde{y}) + (r + \lambda)\kappa \tag{10}$$

That is, the hiring productivity threshold is higher than the firing productivity for the marginal worker with a factor directly dependent on κ. This is key in the model: while higher firing costs reduce separation risks for incumbent workers, it spills over onto the hiring margin such that firms become more selective in their hiring practices. This leads to a conditional expected productivity at match formation:

$$y^e := \int_{\tilde{y}} y dF(y) \tag{11}$$

The implication is further that there are three distinct zones in the productivity distribution, as shown in Figure 1. For draws of $y < \tilde{y}$, no matches are formed as surplus is non-positive, which is true globally. The region $y \in \Big(x(y_0), \tilde{y}\Big)$ consists of productivities where a worker wouldn't be hired, but due to the separation cost will not be fired either. Finally, the region $y > \tilde{y}$ consists of matches where flow surplus is positive for all actors such that the match continues. Note again that the cutoff $x(y_0)$ is inherently match-specific.

Using the free entry condition, it is possible to collapse the acceptance rate and ex ante match value for firms into:

$$\frac{c}{q(\theta)} = \int_{\tilde{y}} \Big[\frac{y - x(y)}{r + \lambda} - \kappa\Big] dF(y) \tag{12}$$

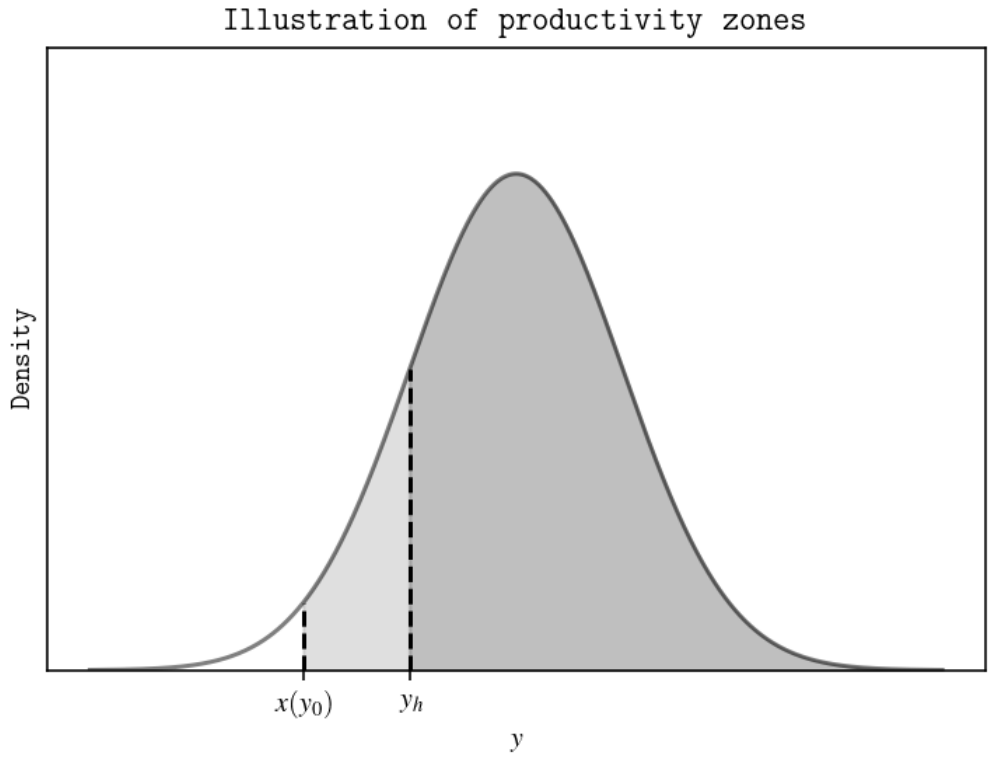


Figure 1: Illustration of productivity zones for a given value of y_0. After a productivity shock, y lands in one of three zones. If $y < x(y_0)$, the firm unilaterally destroys the match. If $y \in [x(y_0), y_h]$ the match continues but the firm experiences negative flow surplus. If $y > y_h$, both parties enjoy positive flow surplus. As wages are fixed, the worker always has positive flow surplus.

This can be plugged into the unemployment flow value equation 4 using the Nash bargaining protocol to find the expression for the reservation wage and close the match-level equilibrium:

$$rU = b + \frac{\beta}{1-\beta} c\theta \tag{13}$$

3.3 Comparative statics

While there are few closed form solutions, it is possible to characterize preferences for tradeoffs, focusing on the jobs **already filled**. Local actors act as in partial equilibrium in that they don't take into account the spillovers values of w and κ have on the market as a whole. In other words, they do not internalize the impact on U or θ when changing

local parameters. I here focus firstly on worker preferences over w and κ. Looking at the derivatives of their value function with regards to these parameters yield:

$$\frac{dW}{d\kappa} = \frac{\lambda(rU - w)x'_\kappa g(x)}{\Big(r + \lambda G(x)\Big)^2} \tag{14}$$

$$\frac{dW}{dw} = \frac{\lambda(rU - w)x'_w g(x) + \lambda G(x) + r}{\Big(r + \lambda G(x)\Big)^2} \tag{15}$$

Note that the implicit derivatives of (7) yield $x'_\kappa < 0$ and $x'_w > 0$. The takeaway is then that workers already in a job monotonically prefer a high κ as they will be able to keep the job for longer, even when their productivity drops. This is trivial as the wages are fixed and they therefore by definition get a positive flow surplus, which they would prefer doing infinitely. The real interesting part is how this is valued in terms of wages. Allowing wages to be flexible, it is possible to characterize the potential tradeoff through the marginal rate of substitution:

$$MRS_{\kappa,w} := \frac{\frac{dW}{d\kappa}}{\frac{dW}{dw}} = \frac{\lambda(w - rU)x'_\kappa g(x)}{\lambda(w - rU)x'_w g(x) - \lambda G(x) - r} \tag{16}$$

Along an indifference curve, the MRS measures how many units of w the worker would give up to get a unit of κ. The key part is that this value in steady state is a function of θ:

$$\frac{\partial MRS_{\kappa,w}}{\partial \theta} = -\frac{c\frac{\beta}{1-\beta}r\frac{\Big(1-\frac{\lambda}{r+\lambda}(1-G(x)\Big)(\lambda G(x)+r)}{\lambda g(x)}}{\left(w - b - \frac{\beta}{1-\beta}c\theta - \frac{\Big(1-\frac{\lambda}{r+\lambda}(1-G(x)\Big)(\lambda G(x)+r)}{\lambda g(x)}\right)^2} < 0 \tag{17}$$

While convoluted, the sign is clear as the numerator consists of clearly positive factors, whereas the denominator is squared. The negative value is one of the key takeaways from the model. When the outside option of unemployment increases through higher tightness and therefore higher job finding rates, workers are less willing to give up money to protect their employment. Vice versa, decreases in tightness, e.g. through a shock to the unemployment rate, makes workers more willing to give up wages to hinder separations.

This is the key mechanism I will investigate in the empirical section, as the implication is that the workers put higher emphasis on employment protection when times are tough on the local labor market.

3.4 Central bargaining

On a central level, however, actors take into account the effects wages and employment protection have on the entire economy. Specifically, the central union is able to internalize the effects of within-job gains on overall employment. It is e.g. trivial that while employed workers would prefer infinite job protection, a central union would not as they would internalize the effect such a setup would have on hiring, which would tend to zero. In other terms, I posit that a central union cares about all the workers within their sector, not only the ones currently employed. To relax the assumption, I allow them to put heterogeneous weights on employed and unemployed workers, which is further reinforced by natural weights stemming from steady-state employment levels. Specifically, I specify the central union's target function as:

$$\Omega = \overline{u}U + \alpha(1 - \overline{u})W^e \tag{18}$$

where $\overline{u}$ is the steady-state unemployment rate and α is weighting factor such that $\alpha > 1$ would imply overweighting employed utility and vice versa. Already here is visible a distinction to the local level: central actors have to factor in changes in $\overline{u}$ stemming from changes in κ. The unemployment dynamics can be written as:

$$\dot{u} = \lambda \int_{\tilde{y}} G(x(y_0))n(y_0)dy_0 - up(\theta)[1 - G(\tilde{y})], \tag{19}$$

where notably the inflow into unemployment depends on the mass of workers starting at different productivities, $n(y_0)$. Normalizing the aggregate inflow such that $\tilde{\sigma} := \lambda \int_{\tilde{y}} G(x(y_0)) \frac{n(y_0)}{1-u} dy_0$ and setting $\dot{u} = 0$ yields the simple expression:

$$\overline{u} = \frac{\tilde{\sigma}}{\tilde{\sigma} + a(\tilde{y})p(\theta)} \tag{20}$$

This yields a hump-shaped relationship between the steady state unemployment rate and the firing cost κ, as taking total derivatives yield:

$$\text{sign}\frac{d\overline{u}}{d\kappa} = \text{sign}\left[\frac{\partial\tilde{\sigma}}{\partial\kappa} - \tilde{\sigma}\left[\frac{\frac{\partial a(\tilde{y})}{\partial\kappa}}{a(\tilde{y})} + \frac{\frac{\partial p(\theta)}{\partial\kappa}}{p(\theta)}\right]\right] \tag{21}$$

The first term is strictly negative, as shown by (7). Increases in κ changes the firing productivity threshold, such that the firms are less willing to destroy matches due to productivity fluctuations. The second composite term is strictly positive: increases in κ affects the hiring threshold and therefore acceptance rate through (10). Similarly, it affects vacancy creation by reducing ex ante match value on the firm side, as shown in (12). Evaluating this expression at $\kappa \to 0$ yields a standard DMP with wage rigidity, with dominance of the first term. Similarly, evaluating $\kappa \to \infty$ yields $a, p \to 0$ with dominance of the second term. This naturally implies the existence of a finite κ such that $\frac{d\overline{u}}{d\kappa} = 0$.

It is thus the case that even if a central union with a target function given by (18) could choose freely, they would set a finite κ. However, they rarely choose these values freely in real life. To discipline κ, I let it be a bargaining object between the central union and a central employer confederation with target function $(1-\overline{u})J^e$. The central employer thus cares about the quantity and quality of filled matches. I again use a Nash bargaining protocol with central union weight γ, yielding the following condition for the centrally supplied κ_c^*:

$$\kappa_c^* = \arg\max_{\kappa}(1-\overline{u})\Big[\alpha W^e - U\Big]^{\gamma}\Big[J^e\Big]^{1-\gamma} \tag{22}$$

$$\implies \frac{u_\kappa}{1-\overline{u}} = \gamma\frac{\alpha W_\kappa^e - U_\kappa}{\alpha W^e - U} + (1-\gamma)\frac{J_\kappa^e}{J^e} \tag{23}$$

At $\kappa \to 0$, the positive effects of increases in reducing the unemployment rate dominate the negative effects on expected surplus. On the other hand, $\kappa \to \infty$ reduces job creation to the extent that the negative effects dominate. There thus exists an interior solution, and the firing cost κ_c^* fulfilling this criterion is positive and finite. With the match-level bargaining, this implies an allocation of wages and bargaining $\Big\{w(\kappa_c^*), \kappa_c^*\Big\}$.

3.5 Match-level trade

While the allocations stemming from central bargaining and subsequent match-level wage bargaining are optimal with regards to the central joint maximization problem, they are not necessarily optimal for the local actors. That is, because of differences in constituency, scope and timing, it might be possible for the local actors to reach Pareto improvements from the centrally supplied allocation during an ongoing match. The institutional setting allows for exactly this: local parties can circumvent CBA:s at their choosing conditional on having a local union club, and this is a unilateral decision on the worker part. Specifically, I model this as an alternative tradeoff in the (w, κ)-domain, the wage-employment security tradeoff. This can be thought of as trade with an initial allocation such that κ_c^* is centrally supplied and w set through (9). As long as the actors' marginal rates of substitutions are not equalized, there are Pareto improvements to be had through moving to another allocation (w_P^*, κ_P^*). By taking derivatives of $J(y)$ with regards to w and κ, I construct a marginal rate of substitution for the employer side as well, which simply equals $G(x)\lambda$. Equaling this expression to the incumbent worker MRS yields a condition for local Pareto optimality:

$$W(w, \kappa) - U = \frac{G(x)}{g(x)(r + \lambda)} \tag{24}$$

Note that the allocation $\left\{w(\kappa_c^*), \kappa_c^*\right\}$ does not in general lie on this contract curve, creating scope for local trade. Specifically, there are wedges between the ex ante bargaining and the match-level optima stemming from internalization of externalities, optimization of expected values instead of realized ones, as well as the relative weighting of the states on the central union side. As hinted at by the worker MRS, the relative gains from trade are further functions of macroeconomic parameters. The worker's willingness to trade away wages for κ is a direct function of θ: the same is not true for firms as a result of the free entry condition.

Thus, if the local actors were allowed to freely trade from the initial allocation $\left\{w(\kappa_c^*), \kappa_c^*\right\}$, they would in general do so. The direction of trade would then depend chiefly on two parameters: layoff density $g(x(y_0))$, labor market tightness θ. In situations where the

labor market is tight or the firing risk is low, there are Pareto improvements to be had where the workers trade away employment protection for increased wages. On the other hand, with high layoff risk or a slack labor market, the local actors would like to trade away wages for κ. In this paper I mainly consider the second case, but it is a promising avenue for future research to consider how workers in a booming economy would prioritize wage growth at the expense of job stability.

3.6 Demand for local unionization

I here consider a labor market with high unemployment risk such that $\kappa_c^* < \kappa_P^*$ and $w(\kappa_c^*) > w_P$, where κ_P and w_P satisfy (24). For tractability I consider trade at constant prices such that $\{w_P, \kappa_P\}$ constitutes a single fixed point and not the set of Pareto optimal points attainable from the initial allocation.

Moreover, in order to keep the model tractable I abstract from having mass at the workplaces. This naturally leads to the concept of union density being abstract, as it in reality is a continuum on the unit interval, whereas it in the model so far only can take the values 0 and 1. In order to be able to use the model to say something about the empirical intensive margin, I therefore assume that the expected realization in the (w, κ)-domain is a weighted average of the local and central allocations, with weights decided by density. This is a simplification of the institutional fact that clubs may choose whether or not they want to assume the bargaining mandate. The reasoning is based on an assumption that a club representing an infinitesimal share of workers probabilistically will not exercise this option, whereas a club covering all workers would. By assumption, the expected outcome is then a convex combination of the equilibria, with weights decided by density η:

$$\begin{pmatrix} w_l \\ p_l \end{pmatrix} = \eta \begin{pmatrix} w_P \\ \kappa_P \end{pmatrix} + (1-\eta) \begin{pmatrix} w(\kappa_c) \\ \kappa_c \end{pmatrix} \tag{25}$$

The resulting allocation is illustrated in Figure 2. Determining η requires imposing a cost on organizing, which I denote by $h(\eta)$. This is due to the fact that movement from $(w(\kappa_c^*), \kappa_c)$ to (w_P, κ_P) is by construction a series of Pareto improvements, and thus

costless trade would lead to $\eta = 1$ and the fully unionized equilibrium. Organizing can however be time consuming and reduce utility by itself, depending on the worker and workplace characteristics and preferences. Moreover, it requires paying union dues which comes at a strictly monetary cost. I assume this cost to be increasing with η in order to capture the heterogeneity in preferences within a workplace: the first worker to organize will naturally have the lowest net cost of doing so, and the last one will have the highest. As the benefits are shared, the logic translates directly to gross cost, i.e. $h''(\eta) \geq 0$, which further means that the function is invertible. Naturally, η will then in equilibrium constitute the point where marginal gains equals marginal costs, if the solution is internal:

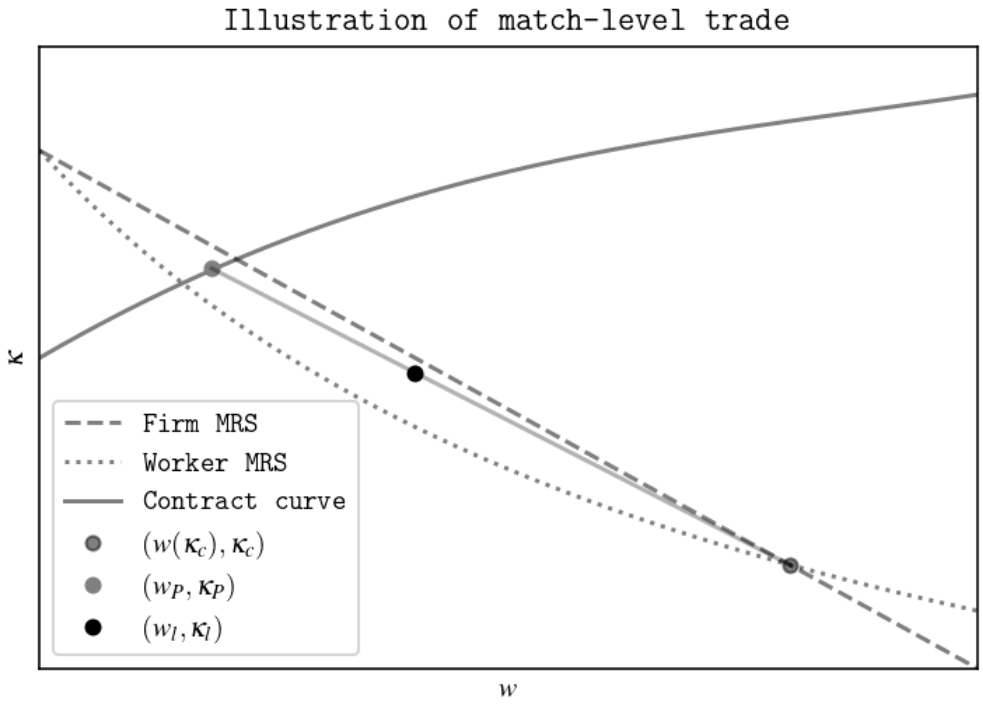


Figure 2: Illustration of match-level trade. The initial allocation $(w(\kappa_c), \kappa_c)$ is the result of the ex ante central bargaining. The MRS are derived from the incumbent value functions. The contract curve satisfies (24). The ex post allocation constitutes the convex combination of allocations according to (25), with endogenous weights depending on macroeconomic conditions.

$$\eta^* = \arg\max_{\eta} W\Big(w(\eta), \kappa(\eta)\Big) - h(\eta) \tag{26}$$

$$\implies h'(\eta^*) = \frac{dW\Big(w(\eta^*), \kappa(\eta^*)\Big)}{d\eta} \tag{27}$$

$$\implies \eta^* = h'^{-1}\left(\frac{\lambda(rU - w)g(x)x_w[w_\eta - r\kappa_\eta] + w_\eta(r + \lambda G(x))}{(r + \lambda G(x))^2}\right) \tag{28}$$

While closed form, it is possible to see that the marginal gains from unionizing is a direct function of labor market tightness:

$$\frac{d\frac{dW\Big(w(\eta^*),\kappa(\eta^*)\Big)}{d\eta}}{d\theta} = \frac{\lambda r g(x) x_w[w_\eta - r\kappa_\eta]}{(r + \lambda G(x))^2}\frac{\partial U}{\partial \theta} \tag{29}$$

In a labor market with high unemployment risk and constant trade prices, it is unambiguously the case that $w_\eta < 0$ and $\kappa_\eta \propto -w_\eta$. Together with the straightforward result that U is increasing in θ, it implies that the marginal gains from increasing union density is decreasing in labor market tightness. As the cost function is invariant to θ, it clearly implies that:

$$\frac{d\eta^*}{d\theta} < 0 \tag{30}$$

The intuition is straightforward - in a labor market with high separation risk, the centrally supplied allocation is not Pareto optimal at the match level as both parties have incentives to trade away wages for employment protection. In such a market, a negative shock to the labor market tightness increases gains from such a trade. Thus, if union density governs the extent of trade, union density increases as the outside option turns worse. This is very much in line with an insurance story: the worse the bad state is, the higher willingness to pay for staying in the good state. In this two-tiered union market, the local unionization are able to provide that insurance.

3.7 Takeaways

There are three important takeaways from the model.

Firstly, the employed worker MRS gives the straightforward interpretation that in the tradeoff between wages and employment security, the latter gains importance when times are tough for the unemployed. This is intuitive: if e.g. job-finding rates or unemployment benefits are low, the value of keeping a job increases. While wages also increase the distance in value between the W and U states, it has a counteracting force in that the threshold for the firm laying you off becomes lower, leading to higher separation rates. Employment security does not have that mechanism if wages are fixed, and value for employed workers is as such monotonically increasing in κ.

Secondly, there exists a difference in preferences for the wage-employment tradeoff between local and central unions. This stems from their different target functions as well as the point at which they evaluate them. For the first point, it is natural to think that a central union would like to maximize the surplus for all workers adhering to their profession. This naturally includes the ones not currently in employment. This can be rationalized by thinking of e.g. maximizing political power: a union that only cares about a subset of its members would quickly lose membership and thus influence. With this argument follows that unemployment rate factors into the central union's bargaining problem. If they were given full allocation mandates, they would thus not let wages and EPL tend to infinity, as that would reduce the employment rate to zero, as well as reduce the value of unemployment. That is, the central union's target function is neither monotonically increasing in w nor κ due to spillovers on $\overline{u}$ and U. While the local union partly shares the non-monotonicity issue with regards to w, they would always prefer a higher κ ceteris paribus. For the second point, there is a key difference in that the central union maximizes over ex ante surpluses, whereas local unions are made up of workers who already are employed. This reinforces the above mechanism: employed workers put less emphasis on the job entry margins. This speaks directly to the concave hump-shaped relationship between unemployment rate and centralization of bargaining presented in the influential Calmfors and Driffill (1988): for union bargaining to yield minimal unemployment rates, it has to be either fully centralized or fully decentralized.

Thirdly, given that this wedge between central and local preferences exist, there are potential gains from trade to be had at the local level. This is in essence a Coasian argument: if κ is decided centrally and wages through the Nash bargaining protocol at hiring, this constitutes an initial allocation. If this allocation is not Pareto efficient and there is costless trade, actors could move to other points in the $\{w, \kappa\}$-space and both have at least as much surplus. Local unions allow for this through CBA circumvention and decentralized wage bargaining. Theoretically, all workers should utilize this right through forming a club and moving bargaining to the local level. This is in real life not the case, which can be rationalized with club formation and individual organization being associated with a utility cost representing a combination of union dues, time constraints and preferences. Should such a cost exist, the probability of local organization is an decreasing function of unemployment value, such that union density is is negatively associated with labor market tightness.

Taken together, these takeaways generate a clear prediction: as local unions value employment security higher than central unions in bad labor markets, there should be more local organization at a given labor market the lower the job-finding rates for the unemployed.

4 Data and Estimation Strategy

To test above predictions, I utilize rich administrative data, combined with plant-level data collected directly from the unions. To get at the mechanisms at play, I employ a matched Difference-in-Differences strategy where I define treatment through notices of layoffs at *other* plants within the local labor market.

4.1 Worker-level data

The core data used is Swedish linked employee-employer data covering the entire population of firms, plants and workers. For individuals, the key dataset is RAMS, which contains worker-job level information on labor market status and earnings on a yearly level, with several observations per worker if they hold several employments in a given year. These are connected to employer data via plant identifiers. While yearly, the data contains information on which months the earnings apply to, such that a monthly panel can be backed out. As the month data is measured through employer surveys with a lot of noise, I will in main specifications collapse the information to the yearly level. Furthermore, the data contains information on their occupations, classified according to the Swedish SSYK system. The SSYK system is such that an occupation is represented by a four-digit number, which can be subset to choose granularity. Starting in 2022, RAMS is replaced with BAS, a similar dataset which instead measures income directly by month through the tax registry. This allows for more trustworthy granularity in the temporal dimension.

I further observe more detailed observations for a subset of workers through the Wage Structure Statistics (WSS). WSS is a stratified sample on the firm level, and contains information on actual wages and hours, as opposed to just earnings. If further contains information on whether or not workers are blue- or white-collar directly, along with their SSYK code for the occupation. This de facto lets me back out a occupation-based definition of blue/white collar, which matters for union-related outcomes as that distinction matters for which club they would belong to, if any. The sampling is done such that large firms are overrepresented in the private sector, whereas it covers the entire population for public employees.

To complete the labor market histories, I merge registers from the unemployment insurance systems and the PES. This is done to distinguish individuals without earnings between active jobseekers and those not in the labor force. Notably, the UI system requires active jobseeking to be eligible for benefits, such that I know that recipients are actively seeking employment. Adding to this, I through other registries such as LISA am able to observe a lot background characteristics, such as educational attainment, sex, marital status, age, and residential area.

For most specifications, I will collapse all this worker-level information to the plant level, as that's the level where I can measure unionization, as described below. This lets me characterize how workers at plants with clubs differ from those without ones, as well as serve as interaction terms in heterogeneity analyses.

4.2 Employer data

For employer information, the key dataset is FEK, a yearly panel containing information on the plant level, with identifiers for the firm. Key variables here are the number of employees at the plant, and information on the plants' economic performance. Furthermore, the data contains information on which industry the plant is operating in, according to the Swedish SNI system which assigns a five-digit number to each industry. Just as with the SSYK, SNI's first digit denotes a course industry, with increasing granularity as digits increase.

4.3 Notice data

As described in 2.4, all plants laying off at least five employees have to report the notice to the PES. From the PES I thus gather all these notices and get information on which plants aim to lay off workers. A key advantage of this over standard mass layoff designs is that I can exclude voluntary job separations without prespecifying some cutoff in size reduction to claim exogeneity. Moreover, I have the finalized lists of who get displaced in the layoff, which lets me follow workers I know for a fact are displaced due to a firm-side choice rather than a quit or jointly optimal separation. The existence of these bilaterally

inefficient separations speaks to the setup of the model, echoing Blanco et al. (2024) and Carlsson and Westermark (2022).

4.4 Union data

Unionization information is not a part of the standard administrative data as unions are separate entities from the state, and the bargaining is done without governmental involvement. There are exceptions to the rule: throughout a few scattered years there existed tax deductions for union membership fees such that individual-level dummies could be constructed from the tax registry. These are not used here, as they are not suited for longitudinal analysis due to long interruptions, nor have any institutional information regarding existence of local clubs. Instead, I have reached out to a subset of central unions and gotten their data on plant-by-year level representation. I have then merged this data to the above administrative registers with the help of Statistics Sweden. While the data differs some between unions, it lets me construct a plant-level panel of local union representation at the extensive and intensive margins, as it contains information both on the existence of a club and its membership count conditional on existing. Moreover, the data contains information on other types of representation, such as non-club bargaining mandates or safety officers. I have further added data on the firm level regarding if the union has utilized their right to seats at the company board, a right they have if they have a club and at least 25 employees.

The sampling of unions is weighted towards the private manufacturing sector. In Appendix A I describe exactly which unions are participating and to what extend they provide good coverage, but the sampled unions by and large cover the entire manufacturing industry less forestry and some minor subindustries. The data spans all three umbrella unions such that blue- and white collars as well as academics are observable. The main analysis will be done using only the Metalworkers' union (IF Metall). This is done for data quality reasons to ensure that there is longitudinal variation with clearly defined pre- and post-periods, and that there is no survivorship bias stemming from observations being conditional on the plant still existing today.

To make sure that the plants without clubs have the central union outcome as the outside option, I have further gathered information on CBA:s from the organizations Fora and Collectum, who both are joint ventures between the unions and the employer organizations handling CBA-related retirement and insurance provisions. Together, the data from these sources provides the universe of firms attached to CBA:s. A drawback is that the data contains nothing on the actual contents of the agreements. To remedy that, I have gathered information from the Swedish Mediation Office, who yearly publish reports containing short-form versions of the near-universe of CBA:s struck that year. While this does not give me the full CBA contents, it gives me most of the information pertaining to wage formation, and the agreements' degrees of centralization.

4.5 Estimation strategy

Key to estimating the causal effect of unemployment value on local unionization is shutting down the potential reverse causality relationship. In terms of the model, the relation I want to study is $u \rightarrow \eta^*$. However, taking spillovers into account, there will be an effect such that $\eta^* \rightarrow u$ as the local unions by protecting incumbents might hurt the hiring margin. Thus, a raw comparison in levels is biased by incorporating the feedback loop. Instead, I need plausibly exogenous variation in the local unemployment rate with regards to existing local union density.

To do so, I mainly employ a matched difference-in-difference design at the plant level, where I define treatment through notices of layoffs at other plants in the local labor market. The idea is that if there is a layoff at plant A, subsequent unionization decisions among workers at plant B are stemming from their worsened outside options, i.e. $U \downarrow$. The key identifying assumption is that whatever caused the layoff at A is orthogonal to the pre-existing union density at B. Note that this variation will by design capture industry- and occupation-specific shocks. This is not a problem with regards to workers as the effects go through the outside option channel, but might affect employer decisions. Explicitly, e.g. an industry-wide shock inducing a layoff at A might lead to an increased probability of layoffs also at B. This can be seen through the lens of $p \downarrow \rightarrow (J - V) \downarrow$ which increases the willingness of the firm to pay κ. I choose to interpret this as a part of

the mechanism as it naturally increases worker valuation of κ as the logic follows through: decreases in p leads to a reduction in τ^e for any productivity level as it shifts all the level curves upwards, thereby increasing Δ since its derivative is monotonic with regards to τ_c.

In order not to conflate changes in unionization with sectoral trends in organization, I construct a control group using 1:1 matching. The core idea is to for each plant j experiencing a layoff in their local labor market find a control plant within the same industry but at a different location that does not experience such an event. After matching, I go on to estimate an equation of the form:

$$Y_{jtc} = \sum_k \delta_k \times \mathbb{1}(t = c + k) \times Treated_j + \sum_k \gamma_k \times \mathbb{1}(t = c + k) + \pi_t + \alpha_j + \varepsilon_{jtc} \quad (31)$$

The two-way fixed effects model takes into account plant-specific and calendar time-specific differences through α_j and π_t, whereas the coefficients of interest are δ_k: dynamic differences in outcomes between treated and control plants relative to the event year c. Naturally, another core identifying assumption is that the plants would have similar trajectories in the absence of treatment. While it's of course impossible to validate in the post-period due the counterfactual being unobservable, it is possible to investigate pre-trends by checking if $\delta_k \neq 0$ when $k < 0$. There are two main outcomes to consider. Firstly, I calculate the share of union members at the workplace, which constitutes an intensive margin response. Secondly, I investigate the extensive marging through the probability of forming a local union club, at which point the regression constitutes a linear probability model.

5 Results

5.1 Descriptives

5.1.1 Local labor markets

There are several ways to define a local labor market. In my main specification, I denote a local labor market as the intersection between geography at the commuting zone (CZ) level and industry on the 2-digit SNI level. These commuting zones are a many-to-one mapping of municipalities constructed using Statistics Sweden's 2020 definition of local labor markets (SCB 2020), and are based on worker flows. By fixing the CZ definition in time I maintain a consistent unit of observation, with the caveat that the validity of the measure might be time-variant. Focusing on CZ:s as opposed to municipalities allows me to take into account the relevant outside options from a worker perspective. This is especially true in the Stockholm region, which consists of many small municipalities sharing the same labor market. In Northern Sweden, where municipalities tend to be bigger, there are accordingly fewer municipalities per CZ.

As there are 69 commuting zones and 88 2-digit level industries, I start up with 6072 local labor markets. Naturally, not every CZ has every industry, nor do I observe unionization for all industries due to the data collection only covering a subset of sectors. Taking those factors into account reduces the number of effective local labor markets to 788. Using the PES data, I can for each commuting zone calculate the registered and open unemployment rate, and there is substantial variation as shown in Figure 3.

5.1.2 Unionization

Table 1 shows characteristics for plants with observed local union representation after dropping industries where I never observe unionization, again noting that I don't observe the full population of local unions. It is clear that these plants are not randomly drawn: the unionized plants are remarkably larger, with a higher share of natives and men. Interestingly, the unionized plants also consist of workers who are younger and more educated with clearly higher earnings. This latter part is not true when looking at the subset for which we observe direct wages, indicating that there could be differences in

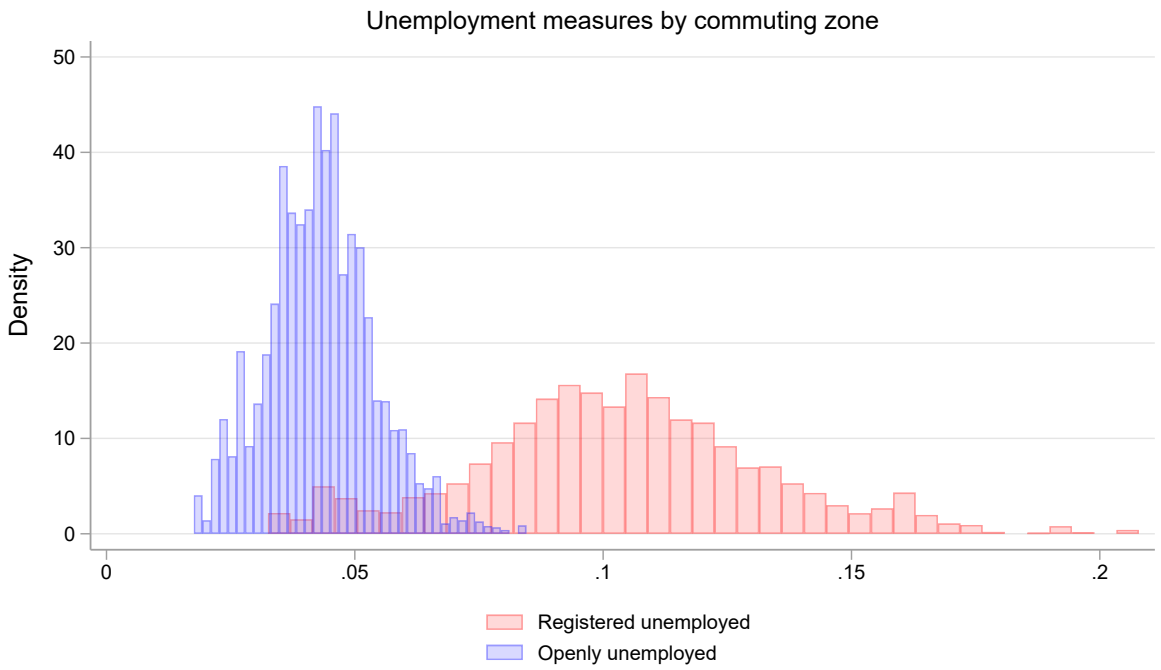


Figure 3: Distribution of unemployment rate by commuting zone. Registered unemployed are registered with the PES and actively seeking employment. Openly unemployed is the subset not participating in active labor market programs.

the intensive margin. Moreover, unionized plants are likely to be situated in local labor markets with marginally higher unemployment rates, as well as significantly higher market concentration, as measured by the sum of squares of employment shares within industry and commuting zone.

There is further dispersion at the intensive margin. I define union density as the number of union members over the size of the plant. Figure 4 shows a histogram of this density, which has a mean of 56% but substantial variation. As for external validity, this number is not to be seen as a representation of the economy as a whole: aggregate numbers suggest that while unionization is higher in the manufacturing sector *within* the blue-collar sector, it is also the case that white-collar unionization is higher than blue-collar. Moreover, these densities are on the plant level as opposed to the individual, which might mask differences over the plant size distribution. Within the sample, density is negatively correlated with density, as shown in Figure 5. A potential explanation for this phenomenon is that larger plants tend to employ a broader set of occupations, who may not be covered by the union data.

Table 1: Comparison of observable characteristics

	(1)		(2)		(1)-(2)	
	0		1		Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Age	454561	46.542	16757	43.520	471318	3.022***
		(0.017)		(0.031)		
Earnings	454565	2103.381	16757	3637.817	471322	-1534.436***
		(3.147)		(6.599)		
Native	454565	0.811	16757	0.844	471322	-0.033***
		(0.000)		(0.001)		
University	454565	0.159	16757	0.202	471322	-0.043***
		(0.000)		(0.001)		
High school	454565	0.589	16757	0.635	471322	-0.045***
		(0.001)		(0.001)		
Less than high school	454565	0.252	16757	0.163	471322	0.088***
		(0.000)		(0.001)		
Female	454561	0.136	16757	0.189	471318	-0.054***
		(0.000)		(0.001)		
Wage	32156	34386.428	7663	33833.399	39819	553.029***
		(79.678)		(67.667)		
Plant size	454565	9.388	16757	181.249	471322	-171.861***
		(0.087)		(4.961)		
CZ unemployment rate	371124	0.090	13223	0.098	384347	-0.007***
		(0.000)		(0.000)		
HHI	454565	0.081	16757	0.179	471322	-0.099***
		(0.000)		(0.002)		

*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

The table shows a raw comparison between plants with and without observed union representation, with unionized plants in column 2. Worker characteristics are means at plant level. Wages are only observable for a subset of plants, see 4.1. Only metalworker union included.

In the cross-section, density further exhibits a positive correlation with the commuting zone unemployment rate, as shown in Figure 6. This is consistent with the modeling predictions as high levels of U increases the relative value of local unionization, but it is by no means causal as there could be bias from reverse causality and/or omitted variables. Again, this is the motivation for shocking the value of unemployment locally as to get rid of endogeneity issues.

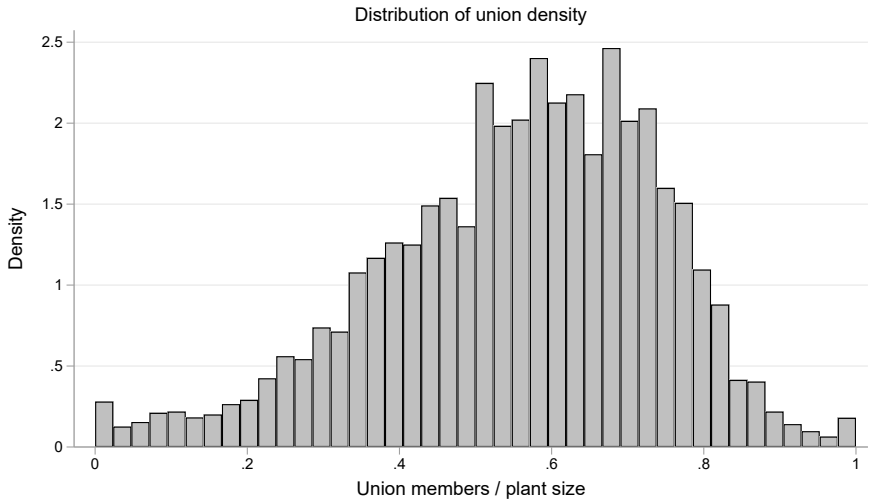


Figure 4: Distribution of union density. Union density is defined as the ratio of union members to plant size. Only metalworker union included.

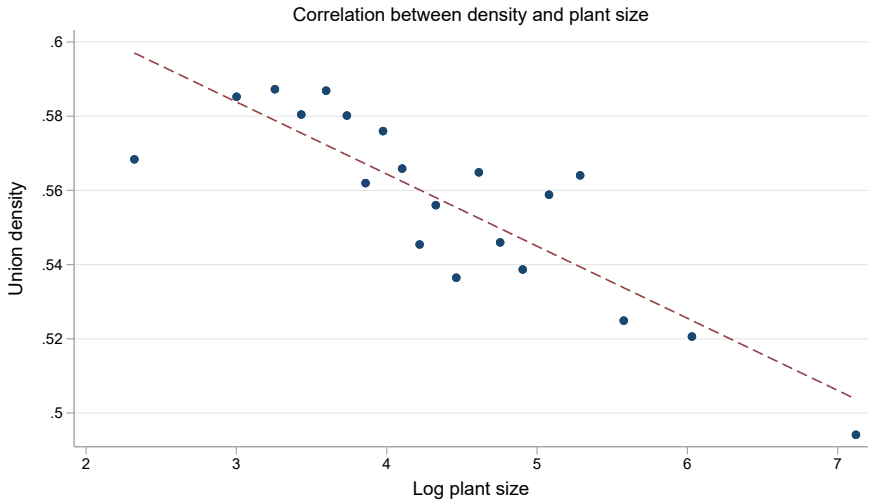


Figure 5: Binned scatterplot of log plant size and union density, with the latter defined as the ratio of union members to plant size. Only metalworker union included.

5.1.3 Layoffs

To define treatment, I consider notices reported to the PES as shocks to the local labor market and assume them to be orthogonal to the previous unionization status of other firms within the same industry-commuting zone cell. Without utilizing the union data, the notice characteristics are described in Figure 7. The modes paint a negative picture: the most common occurrence is for the plant noticing all workers and for all noticed

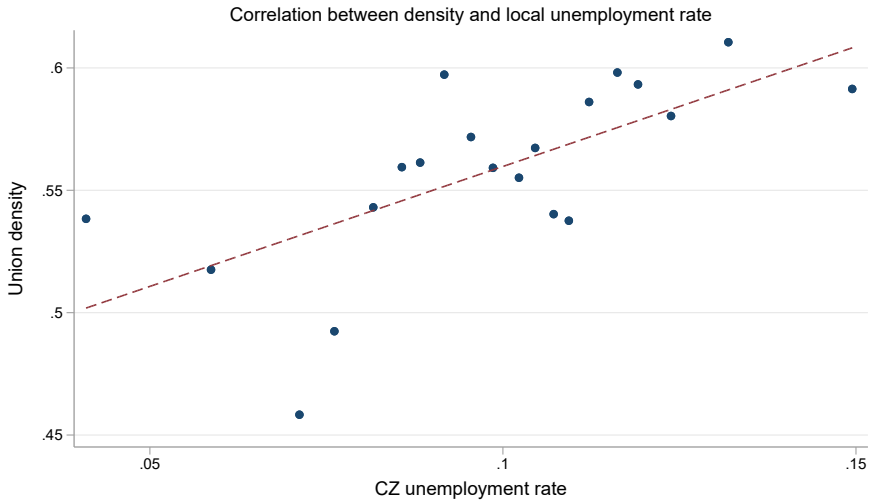


Figure 6: Binned scatterplot of commuting zone unemployment rate and union density, with the latter defined as the ratio of union members to plant size. Only metalworker union included.

workers to become displaced. As shown in Table 2, it is however clearly the case that union presence is negatively associated with layoff severity. While not the direct focus of this paper, it is cross-sectionally the case that union presence is associated with a reduction in the number of people notified relative to the plant size, and conditional on the notification size a reduction in the final layoff size. That is, layoffs at locally unionized firms are less severe both ex ante and ex post the bargaining process. This is pivotal to the mechanism presented: in order for reductions in unemployment value to spill over to unionization, it is theoretically a prerequisite that local unions can protect employment levels.

While clearly the case descriptively that layoffs are less severe at unionized workplaces, it might still be the case that union density is associated with the frequency of layoffs. That is to say, there could potentially be more separations at unionized plants if layoffs occur more often even if the individual layoff is smaller. In Table 3 I check for this and find that it is unconditionally the case, but that the effect disappears when accounting for industry and commuting zones. Thus, it does not seem to be the case that within a local labor market, union density is driving plants into layoff procedures.

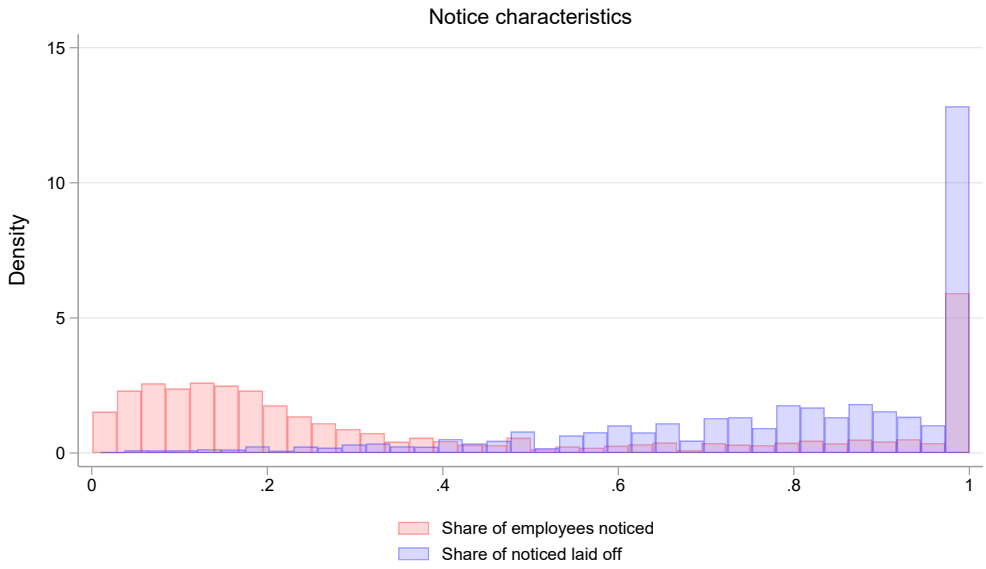


Figure 7: Histogram of notice size. The share of employees noticed are defined by the ration of the ex ante reported size of the layoff over the plant size. The share of noticed laid off is defined by the ex post number of displaced over the ex ante reported layoff size.

Table 2: Correlation between union presence and layoff severity

	(1) Share displaced by notified	(2) Share notified by plant size
Union club	-0.181***	-0.0900***
	(0.013)	(0.008)
Constant	0.443***	0.823***
	(0.006)	(0.004)
N	4255	4174

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The table shows an OLS regression of layoff severity on union presence. Displaced by notified is defined as the ratio between the final layoff size and the initially reported size. Notified by size is defined as the ratio between the initially reported size and the plant size. Union presence is defined as having a club. Only the metalworker union included.

5.1.4 Hiring

In order to rationalize the cross-sectionally higher unemployment rate with lower layoff severity, it must mechanically be the case that hiring is lower in the unionized plants. This would be consistent with the modeling implications that firms adjust their reservation productivities as a response to unionization, leading to a lower hiring rate. Indeed it seems

Table 3: Correlation between union density and layoff frequency

	(1) Layoff	(2) Layoff	(3) Layoff	(4) Layoff
Union Density	0.036***	0.027**	0.013	0.009
	(0.010)	(0.010)	(0.011)	(0.011)
Constant	0.041***	0.046***	0.054***	0.056***
	(0.006)	(0.006)	(0.006)	(0.007)
N	16329	16329	16329	16329
Year FE	No	Yes	Yes	Yes
Industry FE	No	No	Yes	Yes
CZ FE	No	No	No	Yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The table shows OLS regressions of layoff frequency on union density. The outcome is defined as the probability that a plant has a mass layoff in a given year. Density is defined as the ratio of union members to plant size. Only the metalworker union included.

to be the case: in Table 4 I firstly investigate the net hiring as defined by the difference in log plant size between two years, and find a significant negative relation between density and net hiring. This effect, as opposed to the layoff propensity, survives controls for local labor markets, implying that there is less hiring in the net. The coefficient, while similar in magnitude, is not statistically significant when controlling for the plant itself. This is likely a power issue stemming from the fact that within-plant longitudinal union density differences is a magnitude smaller than between-plant variation. Taken together these results imply that if separations are less pronounced, but net hiring is negative, more unionized plants should either exhibit lower gross hiring, or a higher degree of voluntary separations.

Utilizing the rich individual data, I can measure the hiring margin directly. I designate a worker as a new hire if they are employed at firm j in time t but not in $t-1$. I thereafter aggregate to the plant level the proportion of workers that are new hires. The results are shown in Table 4 and robustly imply that plants with higher union density have a lower proportion of new hires. This is again in line with the modeling results that unionization implies higher job security for incumbents at the cost of lower action on the hiring margin.

Table 4: Relation between union density and hiring behaviour

	(1) Net hiring	(2) Net hiring	(3) Net hiring	(4) Net hiring	(5) New hires	(6) New hires	(7) New hires	(8) New hires
Union Density	-0.0389***	-0.0385***	-0.0454***	-0.0461***	-0.0293***	-0.0281***	-0.0338***	-0.0396
	(0.00605)	(0.00300)	(0.00326)	(0.00344)	(0.00823)	(0.00819)	(0.00892)	(0.0216)
Constant	0.201***	0.201***	0.205***	0.205***	0.0149**	0.0142**	0.0173***	0.0209
	(0.00355)	(0.00176)	(0.00190)	(0.00199)	(0.00483)	(0.00481)	(0.00519)	(0.0121)
N	33854	33854	33854	33854	16153	16153	16153	15907
Year FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Industry FE	No	No	Yes	Yes	No	No	Yes	Yes
CZ FE	No	No	No	Yes	No	No	No	Yes

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The table shows OLS regressions of net hiring on union density. Net hiring is defined as the difference in logs in plant size between two years. New hiring is defined as the proportion of workers at a plant who did not work there in the previous year. Density is defined as the ratio of union members to plant size. Only the metalworker union included.

Taken together, the descriptive evidence suggests that unionized plants are associated with a higher local unemployment rate, whereas the entry into unemployment from involuntary separations is lower. While the unemployment figures are cross-sectional, it is mechanically the case that a stable unemployment rate with lower separation rates can only be supported by lower hiring figures. This is clearly the case: taking the coefficients at face value, a movement from the 10th to the 90th percentile in terms of union density is associated with a reduction in the proportion of new hires of about 2.3 percentage points, which is about 11 percent relative to the mean. This is naturally descriptive, as there might be selection into unionization. Specifically, the comparisons might suffer from the reverse causality issues of unemployment rate being a function of unionization density as opposed to the opposite. Nonetheless, all empirical evidence is qualitatively consistent with the comparative statics derived from the model.

5.2 Matching

To create ex ante comparable plants where the treated are exposed to a shock in unemployment value, I employ 1:1 matching as discussed in the previous section. I firstly employ coarsened exact matching such that treated and control firms exactly match on subindustry, year and plant size quartile. I then employ matching within these cells using Mahalanobis' distance on a set of covariates, including worker characteristics and non-coarsened plant size. In order to preserve power I allow for matching with replacement.

The results of the matching are shown in Table 5. I end up with a sample of 1467 plants whom I can properly match to controls, with the treated ones experiencing shocks to their local labor markets. There are some differences in covariates as the treated plants have a marginally smaller share of natives and slightly lower earnings. I deem it improbable that these differences are large enough to drive any subsequent unionization results. For regression results, I further impose that the treated and control plants are visible in the data for the same amount of years, in order not to convolute the coefficients with compositional changes.

Table 5: Results of matching procedure

	(1)		(2)		(1)-(2)	
	0		1		Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Age	1467	43.761	1467	43.486	2934	0.275
		(0.244)		(0.244)		
Earnings	1467	2350.798	1467	2484.514	2934	-133.716***
		(33.514)		(39.345)		
Native	1467	0.882	1467	0.831	2934	0.051***
		(0.006)		(0.007)		
University	1467	0.131	1467	0.147	2934	-0.016*
		(0.006)		(0.006)		
High school	1467	0.625	1467	0.604	2934	0.021*
		(0.008)		(0.008)		
Less than high school	1467	0.244	1467	0.250	2934	-0.006
		(0.007)		(0.007)		
Female	1467	1.150	1467	1.144	2934	0.006
		(0.005)		(0.005)		
Wage	189	30020.660	196	31000.538	385	-979.878*
		(383.251)		(439.872)		
Plant size	1467	26.031	1467	29.009	2934	-2.978
		(2.447)		(3.709)		

The table shows a matched comparison between plants with and without observed union representation, with unionized plants in column 2, evaluated the year before the treatment event. Worker characteristics are means at plant level. Wages are only observable for a subset of plants, see 4.1. Only the metalworker union included.

5.3 Regression analysis

I firstly run a simple two-way fixed effects model without considering matching the controls, to check whether density changes at all around the time of a large mass layoff in the local labor market. That is for plant j experiencing a local labor market shock in year c running:

$$y_{jtc} = \sum_{k \neq -1}^{n} \gamma_k \times \mathbb{1}(t = c + k) + \pi_t + \alpha_k + \varepsilon_{jtc} \tag{32}$$

The main outcome is union density. The results are shown in Figure 8 and, while imprecise, indicate that there seems to be an increase in union density after the negative shock to unemployment value, even after taking into account plant and calendar year fixed effects. Moreover, there are no indications of any pre-trends in density.

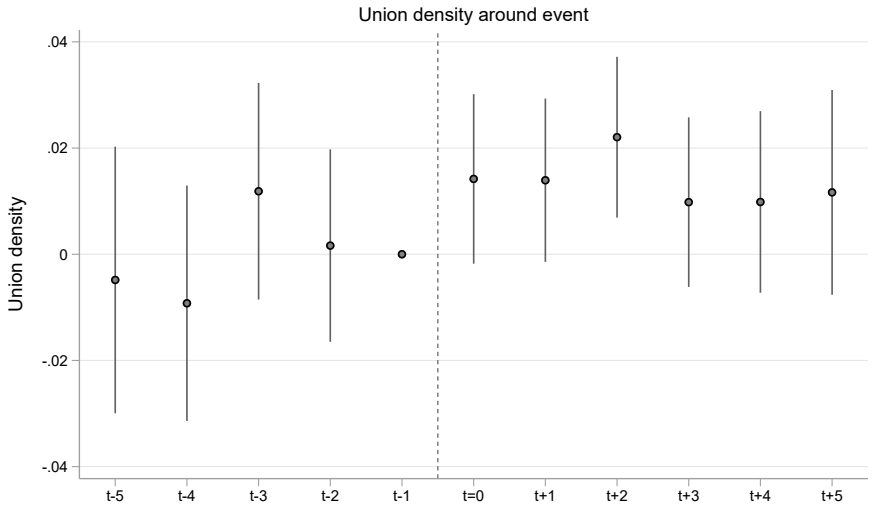


Figure 8: Descriptive event study without matching the control group. The outcome is defined as the ratio of union members to plant size. Only the metalworker union is included.

Going to the matched sample, I instead estimate Equation 31. The main results are shown in Table 6, where I pool the pre- and post-periods in the DiD. As the first column shows, the treatment has meaning in that it affects the local unemployment rate evaluated at the plant level in the direction we would expect. The effect is economically

small: this is to be expected as the treatment is on the commuting zone by subindustry level, and the unemployment rate is measured without taking industry into account. Thus the pass-through of the shock to commuting zone unemployment level is a function of the commuting zone's exposure to the treated industry. We can however clearly see that even on a more aggregate level, the shock is visible.

Table 6: Pooled Difference-in-Difference estimates

	(1) Unemployment rate	(2) Union density	(3) Log members	(4) Union club	(5) Log plant size	(6) Log earnings
Treated-post	0.00356**	0.0145*	0.0978***	0.00451	0.0102	0.00758
	(0.00127)	(0.00636)	(0.0231)	(0.00429)	(0.00879)	(0.00739)
Constant	0.0910***	0.564***	3.852***	0.108***	2.002***	7.901***
	(0.000616)	(0.00341)	(0.0131)	(0.00218)	(0.00399)	(0.00382)
Plant FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	31814	3329	3452	33467	38366	35034

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The table shows pooled difference-in-differences estimates. Treatment is defined as having a large layoff in the local labor market. Unemployment rate is defined as the ratio of registered unemployed to population in the commuting zone. Union density is defined as the ratio of union members to plant size. Standard errors are clustered by commuting zone, 2-digit industry and year. Only the metalworker union included.

In the second column, we see that there is a positive and significant effect of the shock on the intensive margin unionization at the treated plants. That is to say, when there is a negative shock to the local labor market through a layoff, other plants within the same local labor market become more unionized. In the dynamic specification shown in Figure 9, it is visible that the effects are indeed dynamic: it takes a few time periods for the effect to be statistically different from zeroes, after which it remains stable. Expanding the post-period yields qualitatively similar point estimates, but due to the union data only being observed for a limited window, the inference measures quickly become large enough that the minimum detectable effects are improbably large. Moreover, those observations mechanically then have limited pre-periods which introduces questions about the validity of the pre-trends.

As columns 3 and 5 imply, these intensive margin results are more about the numerator than they are about the denominator: the amount of members change substantially whereas the plant size does not. An alternative explanation I then can rule out is that

density increases as the focal plant lays off workers and disproportionally does so for the non-unionized members. A drawback of the data is that I do not observe individual-level variation: I can thus not rule out compositional changes as a whole by for instance splitting the density changes by incumbents and new hires. The lack of negative effects on plant size is further an argument for the employment protection mechanism - it is likely to think that if negative shocks are correlated over commuting zone-industry, all else equal the treated firms should be more likely to reduce their labor usage. This does not look the case, which could be rationalized through the model by the workers at the plants acting on their increased valuation of employment protection.

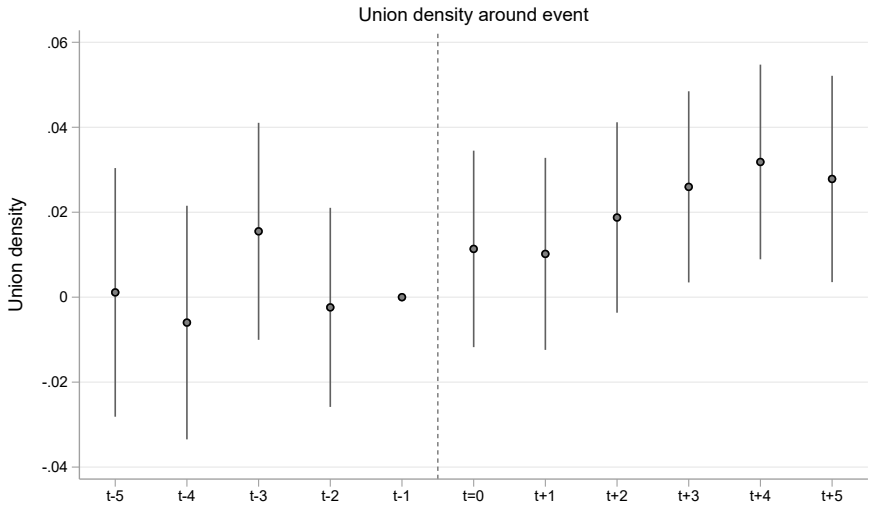


Figure 9: Event study with a matched control. Standard errors are clustered at the CZ-SNI2-Year level. The outcome is defined as the ratio of union members to plant size. Only the metalworker union is included.

The fourth column is a null result and reflects a finding that is there already in the descriptive data: forming a local union club appears to be an absorbing state. That is to say, I very seldom observe clubs closing down or starting up. This has implications for the mechanism at hand: it is institutionally the extensive margin that allows for local circumvention of CBA:s. An explanation that is consistent with the mechanism is however that the intensive margin has implications for the choice to assume the bargaining mandate even conditional on a local union club. The way this is rationalized in the model

is through wage moderation, which does not show up in column 6, as the earnings are calculated for the entire sample, including plants with no observed unionization. When restricting to the sample of compliers, i.e. the treated firms who increase their density following the negative shock to the unemployment value, I observe significant reductions in wages in the post-period when compared to their match control, as shown in Table 7, but this is purely descriptive as the post-shock composition might respond.

Table 7: Pooled DiD for plants increasing union density.

	(1)	(2)	(3)
	Wages	Log wages	Log plant size
Treated-post	-792.6*	-0.0180	0.0209
	(353.9)	(0.00924)	(0.0287)
Constant	35644.2***	10.46***	4.889***
	(166.4)	(0.00459)	(0.0157)
Plant FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	359	359	710

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The table shows regressions for plants increasing their union density after the treatment event. Standard errors are clustered by commuting zone, 2-digit industry and year. Wages are only observable for a subset of plants, see 4.1. Only the metalworker union included.

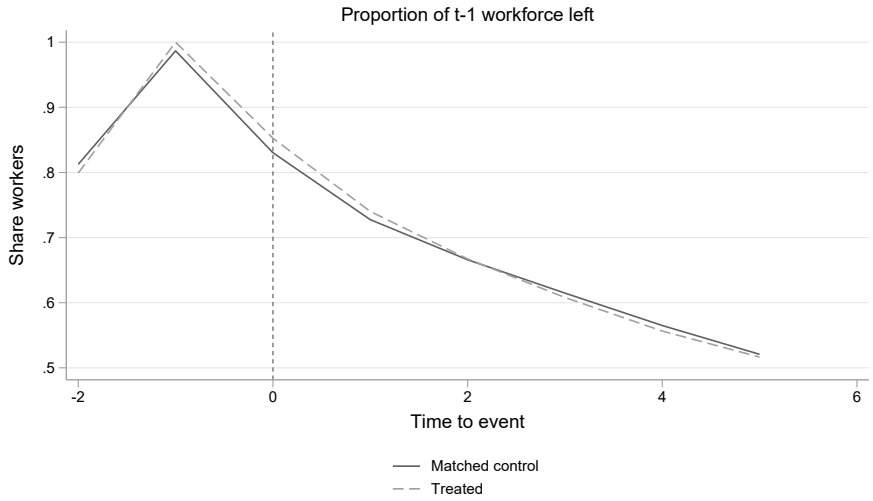
Moreover, I again reject the alternative hypothesis that the plants exhibit higher union density due to discriminatory reductions in their workforces, the key idea being that the treated plants are ex ante more likely to experience e.g. negative product demand shocks. This does not seem to be the case, bu there is an alternative hypothesis that this mechanism is in fact there, but that it is counteracted in the coefficient through the treated plant hiring workers from the noticing firm. It is plausible to think that an increase in the local labor supply through redundancies, which by construction consists of workers relevant for the treated firms, would lead to increased hiring and therefore counteract any shocks on the product demand side. Should these hired workers differ in

their unionization composition, e.g. if the layoff firm was discriminatory in their layoff practices, this could spill over to density estimates.

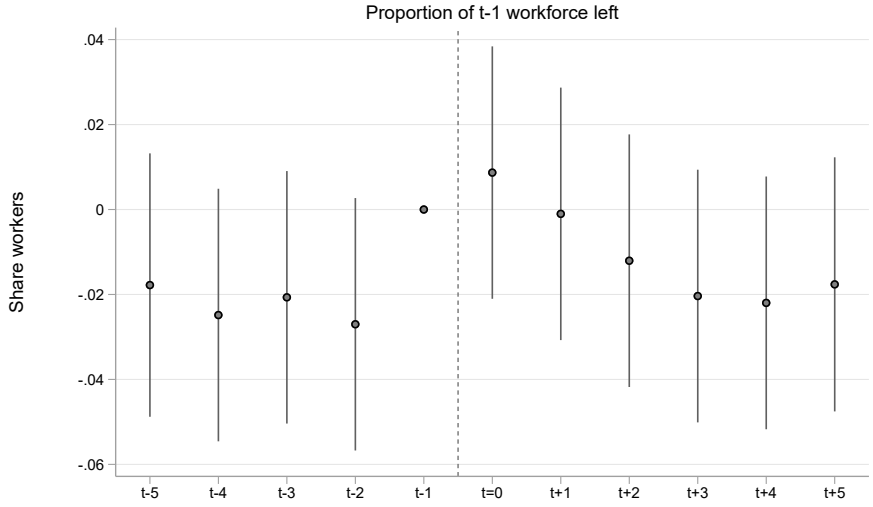
Naturally, the first-best way to look at this would be to have individual-level measures of unionization and observe this directly. As this is not the case, I instead turn to plant size measures split by incumbents and those hired from the layoff firm. That is, I want to rule out that the treated firms have higher churn, laying off incumbent (non-unionized) workers and replacing them with new (unionized) workers from the recently expanded pool of labor. While ex ante implausible, unionization status is likely correlated with other characteristics firms directly observe and (are allowed to) act upon. I investigate this in two ways. Firstly, I fix the pool of workers at $t-1$ and calculate the share of them remaining at the treated and control plants for each year: should the above mechanism be active we would expect negative relative effects. Secondly, I calculate plant size measures net of those coming from the layoff firm, where we also would expect negative effects if the compositional story explains the density coefficients.

The results for the proportion of workers still at the plant, as defined by the composition in $t-1$, are shown in Figure 10 descriptively and through a two-way fixed effect regression. Descriptively, there does not seem to be any large differences between treated and control plants: both have a retention share of 1 in $t-1$ by construction, and thereafter experience similar trajectories, with roughly 70% still remaining at their plants three years later. Accordingly, the regression specification 31 does not pick up any significant treatment effects on retention. The interpretation of these results is that it does not look the case that plants subject to a labor market shock respond by separating from their incumbent workers, thereby changing the composition. As the plant sizes remain similar over treatment status, they must therefore behave similarly in terms of hiring in terms of numbers.

Even though the absolute number of hires have to be similar, there might still be compositional effects stemming from the shock to the local pool of labor supply. Due to LIFO rules, the layoff plants will disproportionally lay off low-tenured workers, changing the distribution of the local labor supply should tenure be correlated with other characteristics such as age and experience. Should the same covariates be correlated with union density, there might be compositional effects on the treated plants as their hires on average have



(a) Descriptive evolution of workforce retention. The outcome is defined as the proportion of workers at the plant in event year $t-1$ still working at the plant in event year t. Treatment is defined through exposure to local labor market shock.

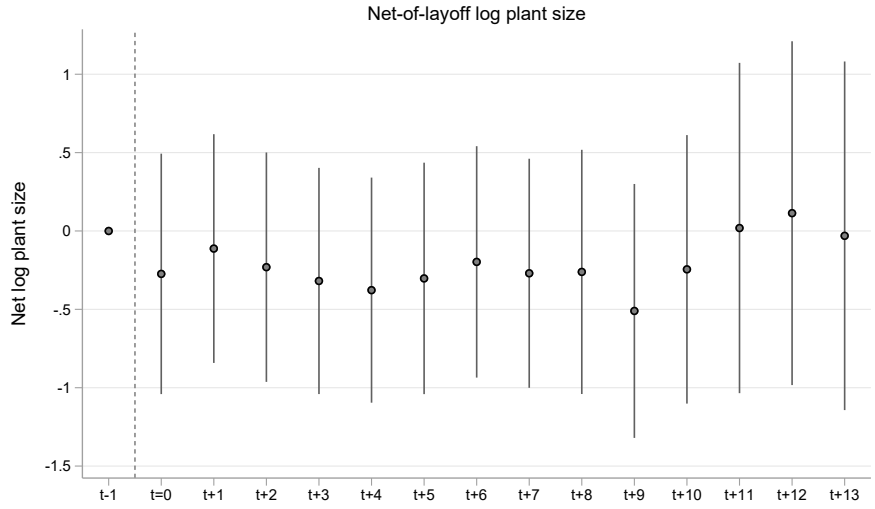


(b) Regression of workforce retention on dynamic treatment. The outcome is defined as the proportion of workers at the plant in event year $t-1$ still working at the plant in event year t. Treatment is defined through exposure to local labor market shock. Standard errors are clustered by CZ-SNI2-Year.

Figure 10: Workforce retention as defined by $t-1$ workforce, by treated and matched control plants.

different unionization status. Indeed, in the model tenure t is positively correlated with the probability to unionize, so all else equal the treated pool of labor supply ought to be less unionized than the counterfactual. While I cannot directly observe this, Figure

11 shows that the plants do not disproportionally adjust their workforces net of workers from noticing plants, nor do they differentially adjust their composition by hiring from these plants.



(a) Regression of log workforce net workers from noticing firms on dynamic treatment.



(b) Regression of proportion of workforce from noticing firms on dynamic treatment.

Figure 11: Characterization of workers coming from layoff plants by treatment status.

6 Conclusion

The question of why workers join the union is by nature dependent on the free riding problem, unless there can be discriminatory practices with regards to employment decisions. In the Swedish context, it is not only the case that such practices are banned by law, but that most labor market outcomes are decided in collective bargaining agreements who by nature cover all workers in a signatory firm, regardless of unionization status. This reinforces the free riding issue: not only is it that at a plant-level, it is prohibited to discriminate depending on union status, but most collectively bargained items are not even decided on the plant level. Yet, people unionize, both as individual members and as participants in plant-level union clubs.

In this paper, I show that the choices can be rationalized by plant-level unions being able to circumvent centrally bargained allocations, and that the preferences on the plant level are fundamentally different from those at the central level. The key idea is that central unions take a macroeconomic perspective and internalize firm responses on the hiring margin, such that the unemployment rate feeds directly into their target function. Building on previous work, I consider the finding that local union clubs are able to lessen the impact of mass layoffs, both in terms of the share of people the firm reports wanting to lay off, and the share of people eventually displaced after the bargaining process. Thus, if local unions are able to protect employment levels, and value them higher than their central counterparts, there is an incentive for the individual worker to become involved in their local union club.

Moreover, this incentive is a direct function of the worker's prospects should they get displaced. In a very tight labor market, displaced workers are likely to find employment quickly again, and as such do not value the local club's emphasis on employment protection. On the other hand, if unemployment rates are high and the workers are likely to experience difficulties in finding new employment, the relative value of maintaining one's current job increases. Thus, if incentives are tied to action, we would expect a positive relationship between local unemployment rates and plant-level club membership.

I take this key prediction to the data I have collected directly from the unions and merged with administrative registers. While the relation is apparent in the cross-section, a

core issue is that the relation might reflect reverse causality. In an insider-outsider sense, it could very much be the case that local union clubs push wages above the perfectly competitive benchmark, and as such unionization would have a causal impact on unemployment rate per se. While this does not look plausible in the cross-section as unionized plants have lower wages, I address this by shocking the unemployment value for plants. I do so by looking at notices of mass layoffs at other plants within the same local labor market, as defined by the intersection of industry and commuting zone. The results are broadly in line with the model predictions: decreases in unemployment value increase unionization on the intensive margin. I do not however find any results on the extensive margin: this is explained by the fact that there is very little within-plant variation in the data. This might be a feature of the sample restrictions as I am predominantly looking at metalworkers in the private sector - results might look different in more dynamic sectors such as those dominated by white-collar workers with more loosely defined professions.

In line with the model, it does look to be the case that the compliers who increase their density as a result of the shock actually do exhibit smaller wage growth as compared to their matched counterparts, but this is purely indicative. A drawback is that I cannot observe the unionization status on the individual level - it would be a natural extension to look at individual-level compliers and their trajectories. That being said, it is plausible to think that the relevant wage measure for keeping employment numbers up is the aggregate one, especially since employers due to LIFO have limited control of who they would lay off should they choose to scale down on their labor usage.

Naturally, the insurance motive through separation frictions presented above only constitutes one part of what makes an individual join a union or start a local chapter. Ideological preferences, workplace culture, amenities, and other local adjustments of collective bargaining provisions are likely to drive workers into membership. Moreover, there are other private gains from membership: among other things, reinforced unemployment insurance is provided with the union membership conditional on unemployment insurance fund membership. This would partially be captured with the empirical design, as if the treatment constitutes an industry or geographical shock such that the worker himself is more likely to be laid off, they would have not only incentives to reduce their probability of being laid off through the firing cost mechanism, but also to increase the value of

unemployment directly through effectively buying a stronger insurance in terms of benefit levels. While unemployment insurance and unions are formally disconnected since 2006-2007, there are many indirect links still through complementary insurances and the occupation-based fund management system. Their interplay, both in terms of take-up and subsequent outcomes, is a promising avenue for future research.

Finally, I think a natural extension would be to consider the relation between the presented mechanism and firm size per se. Allowing firms to have mass, and therefore be heterogeneous, would allow for more nuances with regards to internal union politics (Taschereau-Dumouchel 2020), and monopsony on the labor and product markets (Azkarate-Askasua and Zerecero 2025). It similarly raises questions regarding why firms choose to join collective bargaining agreements, and how their hiring practices are dependent on union organizations even if none of their workers are unionized.

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A Appendix A: Union Data Coverage

Union	Sector	Collar	Members (2025)	Coverage	Club	Delegate
Unionen	Private	White	720,000	2020-, conditional on survival	Yes	Yes
Saco through Sveriges Ingenjörer	Private	White	970,000	2016-2025	Yes	Yes
ST	Public	White	100,000	2016- for clubs, conditional on survival	Yes	Yes
IF Metall	Private	Blue	300,000	2010-2024	Yes	Yes
Handels	Private	Blue	150,000	2017-2024, 2017-2019 no differentiation between delegates (including safety representatives)	Yes	Yes
Kommunal	Public	Blue	500,000	2024 cross-section	Yes	Yes

Table 8: Description of union data

B Appendix C: Supplemental Figures

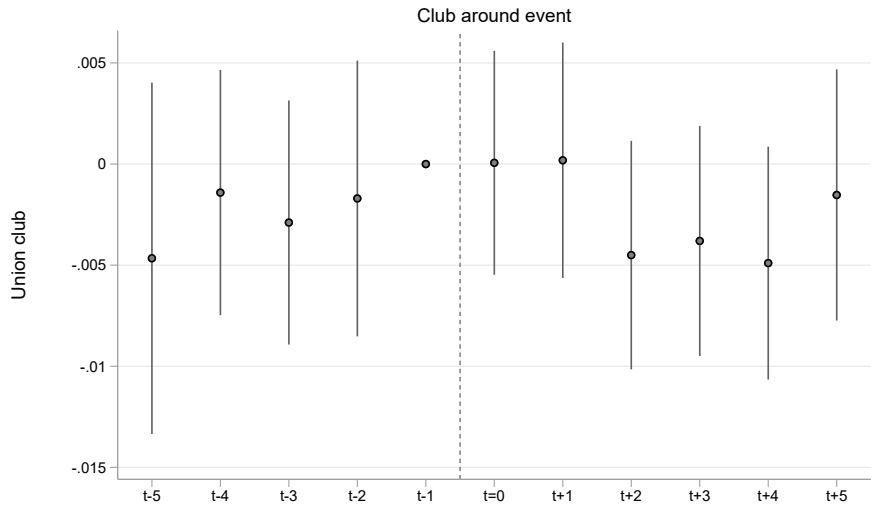


Figure 12: Probability of a new club forming around event. Standard errors clustered on the commuting-zone level.

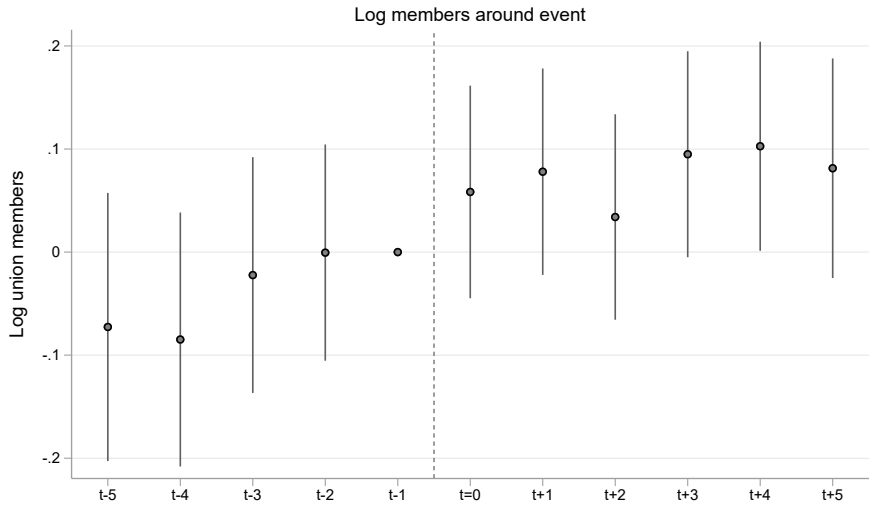


Figure 13: Log number of members in a club forming around event. Standard errors clustered on the commuting-zone level.

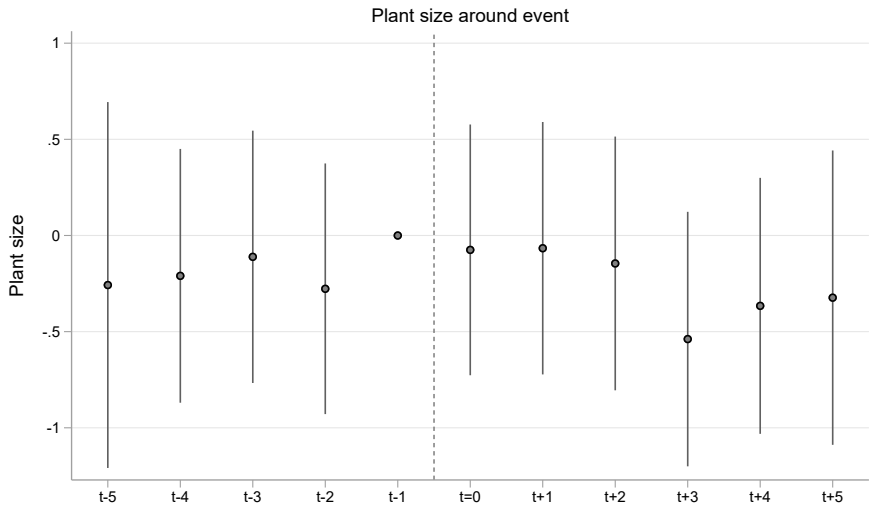


Figure 14: Plant size around event. Standard errors clustered on the commuting-zone level.

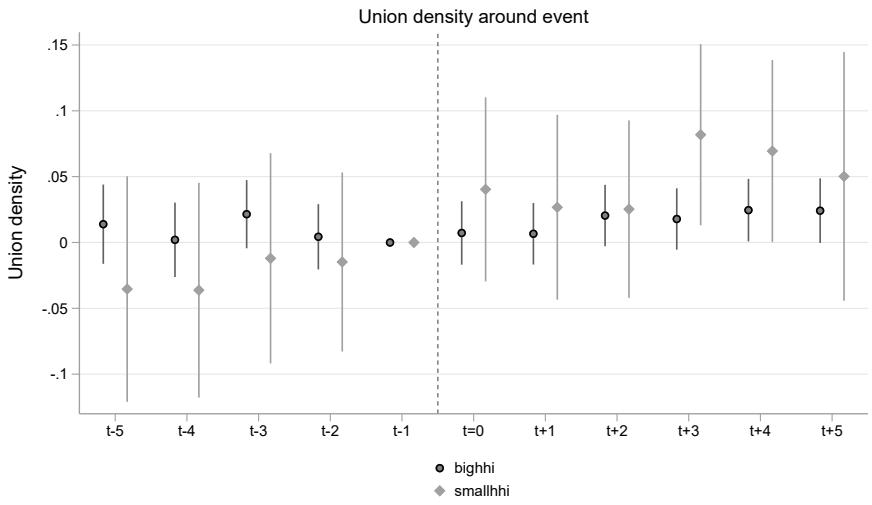


Figure 15: Union density around event, split by market concentration median as determined by the workforce-based measure HHI of the local labor market. Standard errors clustered on the commuting-zone level.